RV College of Engineering

Experiential Learning

Synopsis

2026-27



Title of the Project

Reality Compiler: A Language-Driven Process Verification Engine

Theme: SDG 9

Team No:78

Team Details

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1. Introduction:

Process-based activities are fundamental to organizations, where procedures such as approvals, procurement, maintenance, and quality management are carried out through defined sequences of tasks and decisions. These procedures are commonly communicated using natural-language documents such as standard operating procedures (SOPs), policy documents, emails, and other textual instructions.

Understanding and executing such procedures requires identifying the activities, responsible actors, conditions, decisions, and possible execution paths involved in the process. Converting these descriptions into structured workflow models can make process logic easier to understand, analyse, and verify.

Reality Compiler is proposed as an AI-powered, language-driven process verification engine that bridges natural-language process descriptions and structured workflow models. It uses Large Language Models (LLMs) to extract process elements and organize them into visual workflow graphs, while allowing users to clarify their intended process through interactive questions.

The system further combines graph-based verification, what-if exploration, and source-linked representations to support transparent analysis of the generated workflows. By connecting natural-language understanding with formal workflow analysis, Reality Compiler aims to provide an accessible approach to designing and reviewing reliable process models.

2. Problem Definition:

Problem Statement:

- There is a lack of a unified system that converts natural-language process descriptions into structured workflow models, detects ambiguities and missing cases, verifies process logic, and provides traceable, evidence-based explanations before deployment.
- Organizations run on processes such as approvals, procurement, maintenance and quality signoffs. These are often documented as free-text SOPs, emails or policy notes.
- Free text can be ambiguous or incomplete. Dead ends, missing rejection paths and by-passable approvals may only be discovered after deployment.

3. Objectives:

3.1 Primary Objectives:

- To convert natural-language process descriptions into structured workflow models containing states, transitions, actors, and conditions.
- To identify assumptions, ambiguities, and missing rejection, exception, and failure paths in process descriptions.
- To link extracted workflow elements to their source sentences, providing traceability and transparency.
- To detect dead ends, unreachable states, bypass paths, and invariant violations using graph-based verification.
- To enable interactive Q&A-based workflow generation, producing step-by-step workflows and visual process graphs from natural-language goals.
- To provide what-if exploration and counterexample paths for analysing alternative process executions.

3.2 Secondary Objectives:

- To make process modelling and verification **accessible to non-expert users** through a natural-language interface.
- To provide **evidence-based and explainable verification results** that can be reviewed by process owners.
- To establish a workflow framework that combines **LLM-based process extraction with formal graph analysis**.

4.Methodology

4.1 Approach:

- Natural-language input: The user provides a process description or specifies a desired process through natural language.

- Interactive clarification: The system asks targeted questions when additional information is required to understand the process, conditions, actors, or expected outcomes.
- Process extraction: An LLM analyses the input and extracts key process elements such as activities, actors, states, decisions, conditions, and transitions.
- Workflow generation: The extracted elements are organized into a structured workflow graph representing the process and its possible execution paths.
- Assumption and exception detection: The system identifies assumptions, unspecified cases, and missing rejection, exception, and failure paths.
- Graph-based verification: The generated workflow is analyzed to detect dead ends, unreachable states, bypass paths, and invariant violations.
- What-if exploration: Alternative execution paths and counterexamples are explored to understand how the process behaves under different conditions.
- Source-linked explanation: Each extracted workflow element and verification finding is linked back to the relevant source sentence, providing traceability and evidence for review.

4.2 Workflow:

Natural-Language Process -> Interactive Q&A -> LLM Extraction -> Structured Workflow Graph -> Assumption & Exception Detection -> Graph Verification -> What-if Analysis -> Source-Linked Results

5. Expected Outcome:

- A functional Reality Compiler prototype capable of converting natural-language process descriptions into structured and visual workflow models.
- An interactive Q&A interface that helps users clarify process requirements and generate step-by-step workflows.
- Detection of ambiguous statements, assumptions, missing exception paths, dead ends, unreachable states, and bypass paths in generated workflows.
- Source-linked workflow elements and verification findings, enabling users to trace process logic back to the original description.
- What-if exploration and counterexample paths to help users understand alternative process executions and identify potential issues.

- A quantitative evaluation framework for comparing the proposed system with a plain-LLM baseline in terms of workflow correctness and verification capability.
- A reference architecture demonstrating how LLM-based process extraction can be combined with graph-based verification for accessible and explainable process analysis.

6. SDG 9 Mapping

Sustainable Development Goal 9: Industry, Innovation and Infrastructure

Reality Compiler contributes to SDG 9 by applying AI and process-verification techniques to improve the reliability, accessibility, and efficiency of organizational process infrastructure.

6.1 SDG 9 Targets Addressed

SDG 9.1 – Quality, Reliable and Resilient Infrastructure

- Helps identify process-logic defects such as dead ends, bypass paths, and unreachable states before deployment.
- Supports the development of more reliable and well-structured organizational workflows.

SDG 9.2 – Inclusive and Sustainable Industrialization

- Makes process modelling and verification accessible to non-technical users through natural-language interaction and visual workflow generation.
- Reduces the dependence on specialized process-modelling expertise.

SDG 9.5 – Enhance Research and Technological Capabilities

- Combines Large Language Models, graph-based analysis, and formal verification concepts to create an AI-assisted process-engineering system.
- Encourages innovation in intelligent, explainable, and automated process analysis.